

Vanguard Assurance UX/UI A/B Testing Analysis Report

Executive Summary

This report presents the findings and recommendations from the Vanguard Assurance UX/UI A/B Testing project. The objective of the experiment was to evaluate whether a redesigned digital interface improved user experience and increased process completion rates compared to the previous interface.

The analysis combined data cleaning, exploratory data analysis (EDA), KPI evaluation, hypothesis testing, and Tableau dashboard visualization to assess the effectiveness of the new interface.

The results revealed that the redesigned interface achieved a measurable improvement in user completion rates compared to the previous design. Statistical hypothesis testing confirmed that the observed improvement was statistically significant, meaning the increase was unlikely to have occurred by random chance.

Although the improvement did not fully reach Vanguard's predefined 5% cost-effectiveness threshold, the redesign demonstrated strong positive performance indicators across several key metrics including:

- Higher process completion rates
- Reduced user friction in parts of the user journey
- Improved user engagement patterns
- Better overall navigation flow

The analysis therefore recommends that Vanguard proceed with a controlled rollout of the redesigned interface while continuously monitoring user performance over an additional 60-day optimization period.

1. Introduction

1.1 Background

Vanguard Assurance launched a digital UX/UI redesign experiment to improve the online customer experience. The company aimed to determine whether a more intuitive and modern interface could increase user engagement and improve the completion rate of users navigating through Vanguard's digital process.

The experiment was conducted as an A/B test between:

- Control Group — users interacting with the original interface
- Test Group — users interacting with the redesigned interface

Both groups completed the same sequence of steps:

1. Start Page
2. Step 1
3. Step 2
4. Step 3
5. Confirmation Page

The central business objective was to determine whether the redesigned interface delivered a meaningful and statistically significant improvement in user completion rates.

2. Project Objectives

The primary objectives of this analysis were:

- Analyze user behavior across both interface variations
 - Evaluate completion rates between the Test and Control groups
 - Identify user friction points and error behavior
 - Conduct hypothesis testing to validate statistical significance
 - Assess whether the redesign met Vanguard's 5% improvement threshold
 - Create interactive Tableau dashboards for stakeholder decision-making
 - Provide business recommendations based on data findings
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3. Data Sources

The project used three main datasets:

3.1 Client Profiles Dataset

This dataset contained demographic and customer-related information including:

- Client age
- Gender
- Account balance
- Tenure
- Number of accounts
- Login frequency

3.2 Digital Footprint Dataset

This dataset tracked user interactions throughout the digital process including:

- Process steps
- Visit sessions
- User timestamps
- Navigation patterns

- User progression through the funnel

3.3 Experiment Dataset

This dataset identified whether users belonged to the:

- Test Group
- Control Group

The datasets were cleaned, merged, and transformed using Python before being used for analysis and visualization.

4. Methodology

4.1 Data Cleaning

The following preprocessing steps were performed:

- Removed duplicates
- Standardized column names
- Merged datasets using client identifiers
- Converted date columns into proper datetime formats
- Handled missing values where necessary
- Filtered invalid or incomplete records

Python libraries used included:

- Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Statsmodels
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4.2 Exploratory Data Analysis (EDA)

EDA was conducted to understand:

- User demographics
- Process flow behavior
- Completion patterns
- Error trends
- User engagement metrics

Key observations included:

- Certain process steps experienced higher user drop-offs
- Completion rates differed noticeably between Test and Control groups

- Users in the redesigned interface generally progressed more efficiently through the funnel
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4.3 KPI Analysis

Several KPIs were used to evaluate experiment performance:

Completion Rate

Measured the percentage of users who successfully completed the process.

Error Rate

Measured backward navigation and repeated actions indicating user friction.

Time Spent Per Step

Measured how long users spent on each stage of the process.

User Funnel Progression

Tracked the number of users progressing from one step to another.

User Drop-Off Rate

Measured where users abandoned the process.

5. Hypothesis Testing

5.1 Objective

The goal of hypothesis testing was to determine whether the redesigned interface genuinely improved completion rates or whether the observed difference occurred by random chance.

5.2 Hypotheses

Null Hypothesis (H0)

There is no significant difference in completion rates between the Test and Control groups.

Alternative Hypothesis (H1)

The Test group has a significantly higher completion rate than the Control group.

5.3 Statistical Test Used

A Proportion Z-Test was used because:

- The outcome variable was categorical (completed vs not completed)
 - Two independent groups were being compared
 - The sample size was sufficiently large
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5.4 Results

The analysis showed:

- The Test group achieved a higher completion rate than the Control group
- The observed uplift was approximately 3.71%
- The p-value obtained from the statistical test was below the chosen significance level (alpha = 0.05)

This means the improvement was statistically significant.

Therefore:

✅ The null hypothesis was rejected.

This confirms that the redesigned interface positively impacted completion rates.

6. Cost-Effectiveness Evaluation

Although the redesign achieved statistically significant improvement, Vanguard had predefined a business threshold requiring at least a 5% increase in completion rate for the redesign to be considered fully cost-effective.

6.1 Observed Improvement

Observed completion rate uplift:

- Approximately 3.71%

Required threshold:

- 5%
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6.2 Interpretation

The redesigned interface improved user performance substantially but did not fully meet Vanguard's cost-effectiveness benchmark.

However, the redesign still demonstrated:

- Strong statistical significance
- Positive user behavior improvements
- Reduced friction in parts of the user journey
- Better engagement outcomes

This suggests that the redesign is directionally successful and capable of further optimization.

7. Tableau Dashboard Insights

Interactive dashboards were created in Tableau to communicate findings visually.

Dashboard features included:

- KPI cards
- Completion rate comparisons
- Error rate visualizations
- Funnel progression charts
- Time spent per step analysis
- Demographic filters
- Interactive storytelling components

The dashboards allowed stakeholders to explore performance differences between the Test and Control groups dynamically.

8. Business Insights

Several important business insights emerged from the analysis:

8.1 The New Interface Improved User Completion

Users interacting with the redesigned interface completed the process more frequently than users in the original system.

8.2 Certain Steps Still Introduce Friction

Some process steps showed elevated:

- Backward movement
- Repeated actions
- Longer completion times

This indicates opportunities for further UX refinement.

8.3 Data-Driven UX Design Produces Measurable Results

The experiment demonstrated that data-driven interface optimization can significantly improve user engagement and process efficiency.

9. Final Recommendations

9.1 Controlled Rollout of the Redesigned Interface

The redesigned UX/UI should be gradually rolled out across Vanguard's digital platform because:

- The redesign produced a statistically significant improvement in completion rates
- Users demonstrated improved progression through the process funnel
- The redesigned experience reduced friction in several stages of the customer journey

Although the redesign did not fully meet the 5% cost-effectiveness threshold, the observed 3.71% uplift represents a meaningful improvement over the previous system.

9.2 Implement a 60-Day Optimization Monitoring Period

Vanguard should deploy the redesigned interface under a structured 60-day monitoring and optimization phase.

During this period, Vanguard should:

- Continuously monitor completion rates
- Track user drop-off behavior
- Measure error rates by process step
- Identify bottlenecks in the user journey
- Collect user feedback where possible

The objective of this monitoring period should be to iteratively optimize the interface until the company's 5% improvement threshold is achieved.

9.3 Prioritize Friction Point Optimization

The analysis identified process steps where users:

- Spent excessive time
- Navigated backward frequently
- Repeated interactions unnecessarily

Vanguard should prioritize redesigning these high-friction areas by:

- Simplifying instructions
- Improving button visibility

- Reducing cognitive overload
 - Streamlining navigation
 - Introducing clearer user guidance prompts
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9.4 Introduce User Feedback Mechanisms

Future iterations of the interface should include:

- Optional user satisfaction surveys
- Feedback popups
- Exit intent feedback collection

This qualitative data would complement quantitative metrics and provide deeper insight into user frustrations.

9.5 Continue A/B Testing Culture

The success of this experiment demonstrates the value of continuous experimentation.

Vanguard should establish ongoing A/B testing processes for:

- UX improvements
- Feature releases
- Navigation changes
- Customer onboarding workflows

A data-driven experimentation culture would help Vanguard continuously optimize customer experience and digital performance.

10. Conclusion

The Vanguard UX/UI redesign experiment successfully demonstrated that data-driven interface improvements can positively influence customer behavior.

The redesigned interface achieved:

- Higher completion rates
 - Improved user journey progression
 - Statistically significant performance improvement
 - Better engagement outcomes compared to the original design
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Appendix

Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Statsmodels
- Tableau
- GitHub

Analytical Techniques Used

- Exploratory Data Analysis (EDA)
- KPI Analysis
- A/B Testing
- Hypothesis Testing
- Dashboard Visualization
- Funnel Analysis
- User Journey Analysis

Project Contributors

This project was completed collaboratively by a distributed cross-border team operating across Germany and Spain.

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